LAB 9

Question:

Write a program to implement Floyd Warshall Algorithm to find the shortest path between every pair of the vertices.

C++ Program:

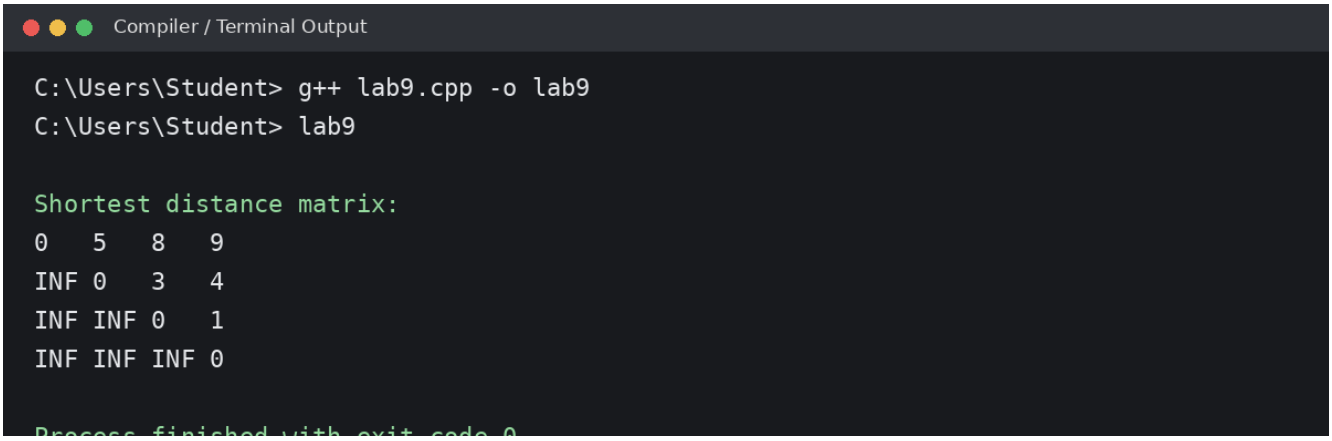
```
#include <iostream>
using namespace std;

const int INF = 1000000;

int main() {
    int n = 4;
    int dist[4][4] = {
        {0,   5, INF, 10},
        {INF, 0,   3, INF},
        {INF, INF, 0,   1},
        {INF, INF, INF, 0}
    };

    for (int k = 0; k < n; k++) {
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (dist[i][k] != INF && dist[k][j] != INF &&
                    dist[i][k] + dist[k][j] < dist[i][j]) {
                    dist[i][j] = dist[i][k] + dist[k][j];
                }
            }
        }
    }

    cout << "Shortest distance matrix:\\n";
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (dist[i][j] == INF)
                cout << "INF ";
            else
                cout << dist[i][j] << "   ";
        }
        cout << endl;
    }

    return 0;
}
```

Output: